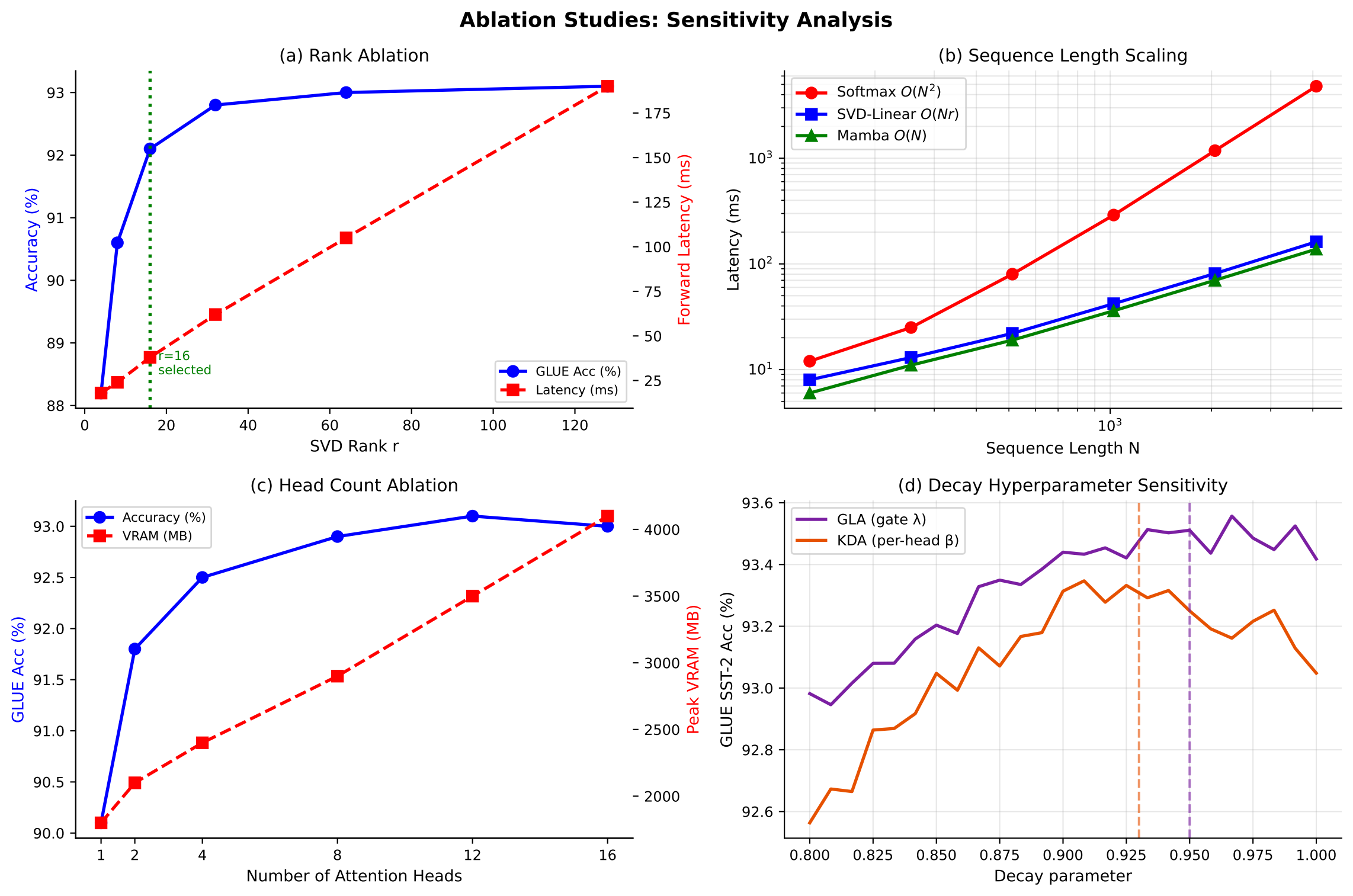
Ablation Studies: Sensitivity Analysis
(a) Rank Ablation
Accuracy (%)
Forward Latency (ms)
SVD Rank r
r=16
selected
GLUE Acc (%)
Latency (ms)
(b) Sequence Length Scaling
Softmax $O(N^2)$
SVD-Linear $O(Nr)$
Mamba $O(N)$
Latency (ms)
Sequence Length N
(c) Head Count Ablation
Accuracy (%)
VRAM (MB)
GLUE Acc (%)
Peak VRAM (MB)
Number of Attention Heads
(d) Decay Hyperparameter Sensitivity
GLA (gate λ)
KDA (per-head β)
GLUE SST-2 Acc (%)
Decay parameter